

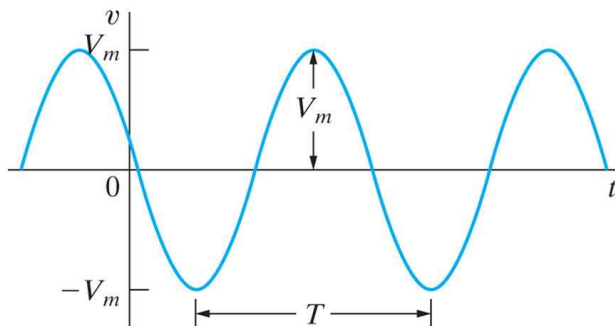
Chap.9 Sinusoidal Steady-State Analysis

§9.0 Why sinusoidal waves are widely used in engineering?

- (i) easy to make
- (ii) many sinusoidal phenomena in the real world
- (iii) easy to consider mathematically
(differentiation of a sinusoidal wave is also a sinusoidal wave,
integration of a sinusoidal wave is also a sinusoidal wave)
- (iv) Fourier Analysis
(a signal can be represented as summation of sinusoidal waves)

§9.1 The Sinusoidal Source

- A sinusoidal voltage



$$v(t) = V_m \cos(\omega t + \phi) = V_m \cos(2\pi f t + \phi)$$

amplitude : V_m [V]

period : T [sec]

frequency : f [Hz] or [1/sec]

angular frequency : $\omega = 2\pi f = \frac{2\pi}{T}$ [rad/sec]

phase angle : ϕ [rad]

- the general equation for a sinusoidal source

voltage source : $v(t) = V_m \cos(\omega t + \phi) = V_m \cos(2\pi f t + \phi)$

current source : $i(t) = I_m \cos(\omega t + \phi) = I_m \cos(2\pi f t + \phi)$

- the rms value of a sinusoidal voltage source

※ rms: square root of the mean value of the squared function
or root-mean-square value

$$\begin{aligned} V_{\text{rms}} &= \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} v^2(t) dt} \\ &= \sqrt{\frac{1}{T} \int_{t_0}^{t_0+T} V_m^2 \cos^2(\omega t + \phi) dt} \\ &= \frac{V_m}{\sqrt{2}} \end{aligned}$$

- ※ The value of AC (alternating current) varies with time.

So we can not represent the voltage of AC as DC (Direct current) voltage.

The effective value (i.e., rms value) of any periodic current is equal to the value of direct current which, flowing through an R ohm resistor, delivers the same average power to the resistor as does the periodic current.

(Example) $v(t) = 300 \cos(120 \pi t + 30^\circ)$ V

a) period

$$\omega = 120 \pi \text{ [rad/sec]}$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{1}{60} = 16.667 \text{ [msec]}$$

b) frequency

$$f = \frac{1}{T} = 60 \text{ [Hz]}$$

c) $v(t)$ at $t = 2.778 \text{ msec}$

$$120 \pi t = 120 \pi \times 2.778 \times 10^{-3} = 1.047 \text{ rad}$$

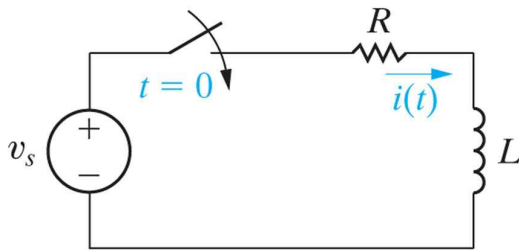
$$= 1.047 \times 180 / \pi = 60^\circ$$

$$\therefore v(2.778 \text{ msec}) = 300 \cos(60^\circ + 30^\circ) = 0 \text{ V}$$

d) root mean square value

$$V_{\text{rms}} = V_m / \sqrt{2} = 300 / \sqrt{2} = 212.13 \text{ V}$$

§9.2 The Sinusoidal Response



$$v_s(t) = V_m \cos(\omega t + \phi)$$

$$L \frac{di(t)}{dt} + Ri(t) = V_m \cos(\omega t + \phi) \quad \dots\dots\dots \text{Eq.(\#)}$$

$$i(t) = \frac{-V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\phi - \theta) e^{-(R/L)t} \quad [\text{transient response}]$$

$$+ \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos(\omega t + \phi - \theta) \quad [\text{steady-state response}]$$

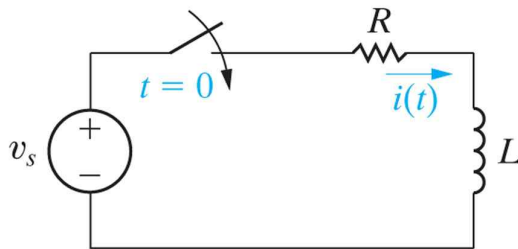
$$\text{where } \theta = \tan^{-1}(\omega L / R)$$

※ It is not easy to solve the second-order nonhomogeneous differential equation (#). [Engineering Mathematics]

● WE FOCUS ON THE STEADY-STATE PORTION IN THIS CHAPTER.

1. The steady-state response is a sinusoidal function.
 2. The frequency, ω , of a sinusoidal response is the same as the frequency of the sinusoidal source driving the circuit.
 3. The amplitude and phase angle of the response are usually different from those of the source.
- ※ The Response of a Linear Systems

● RL circuit with trigonometric function



$$v_s(t) = V_m \cos(\omega t + \phi) \quad \text{Let } i(t) = I_m \cos(\omega t + \theta)$$

$$L \frac{di(t)}{dt} + Ri(t) = V_m \cos(\omega t + \phi)$$

$$L \frac{d}{dt} \{ I_m \cos(\omega t + \theta) \} + RI_m \cos(\omega t + \theta) = V_m \cos(\omega t + \phi)$$

$$-\omega L I_m \sin(\omega t + \theta) + RI_m \cos(\omega t + \theta) = V_m \cos(\omega t + \phi)$$

Now, how to solve ? Easy to solve ?

(※ It takes long time to solve the equation, and we must use many trigonometric identities.)

※ Some Useful Trigonometric Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\cos(\alpha \pm 90^\circ) = \mp \sin \alpha$$

$$\sin(\alpha \pm 90^\circ) = \pm \cos \alpha$$

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha + \beta) + \frac{1}{2} \cos(\alpha - \beta)$$

$$\sin \alpha \sin \beta = \frac{1}{2} \cos(\alpha - \beta) - \frac{1}{2} \cos(\alpha + \beta)$$

$$\sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha + \beta) + \frac{1}{2} \sin(\alpha - \beta)$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1 = 1 - 2 \sin^2 \alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\sin^2 \alpha = \frac{1}{2}(1 - \cos 2\alpha)$$

$$\cos^2 \alpha = \frac{1}{2}(1 + \cos 2\alpha)$$

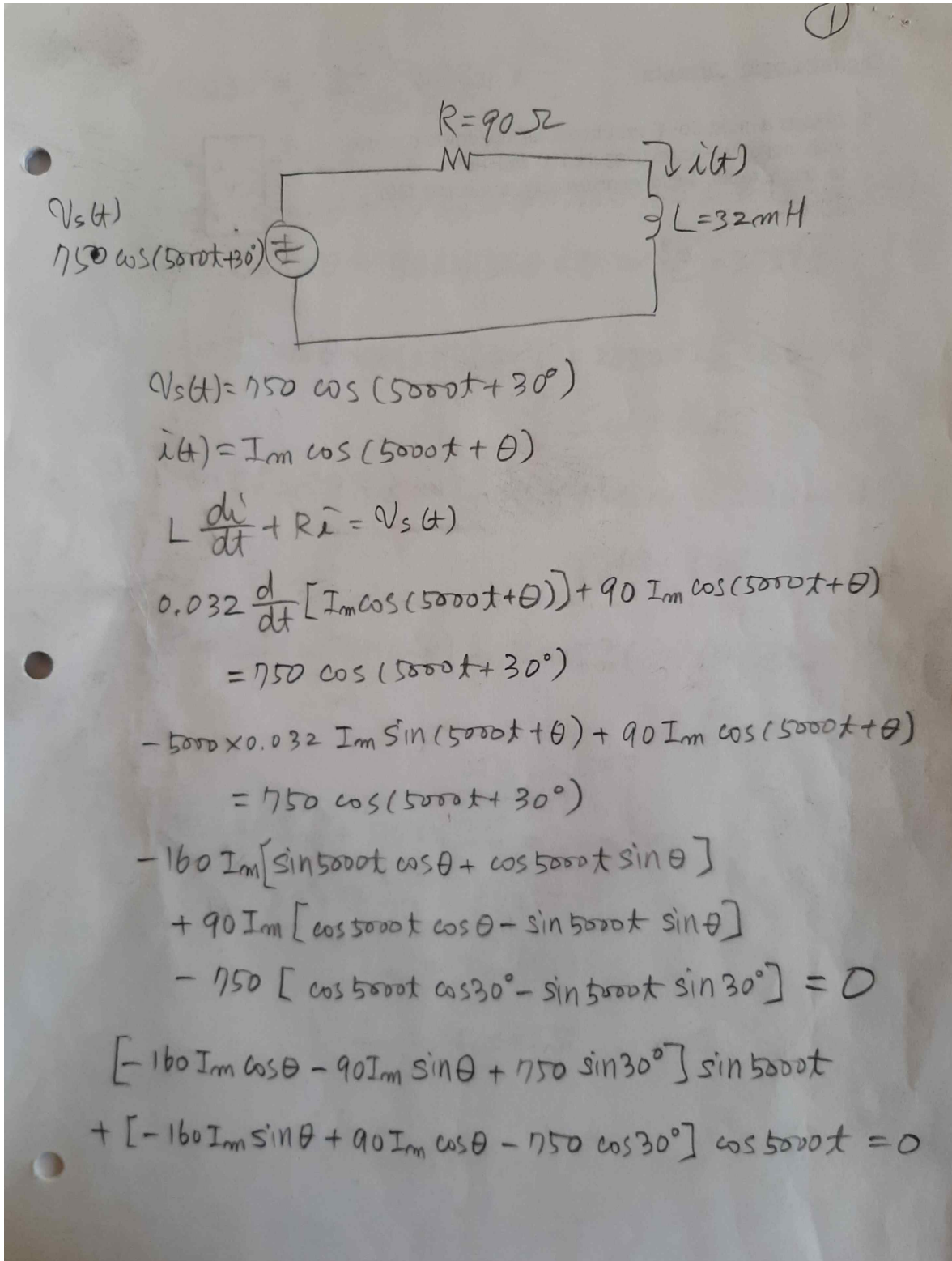
$$\sin \alpha = \frac{e^{j\alpha} - e^{-j\alpha}}{j2}$$

$$\cos \alpha = \frac{e^{j\alpha} + e^{-j\alpha}}{2}$$

$$e^{\pm j\alpha} = \cos \alpha \pm j \sin \alpha$$

$$A \cos \alpha + B \sin \alpha = \sqrt{A^2 + B^2} \cos \left(\alpha + \tan^{-1} \frac{-B}{A} \right)$$

(Example) Find the output signal $i(t)$ using the trigonometric method.



The diagram shows a series RL circuit. A voltage source $V_s(t) = 1750 \cos(5000t + 30^\circ)$ is connected in series with a resistor $R = 90 \Omega$ and an inductor $L = 32 \text{ mH}$. The current $i(t)$ flows clockwise through the circuit.

The derivation follows the trigonometric method:

$$V_s(t) = 1750 \cos(5000t + 30^\circ)$$

$$i(t) = I_m \cos(5000t + \theta)$$

$$L \frac{di}{dt} + Ri = V_s(t)$$

$$0.032 \frac{d}{dt} [I_m \cos(5000t + \theta)] + 90 I_m \cos(5000t + \theta) = 1750 \cos(5000t + 30^\circ)$$

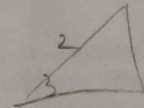
$$-5000 \times 0.032 I_m \sin(5000t + \theta) + 90 I_m \cos(5000t + \theta) = 1750 \cos(5000t + 30^\circ)$$

$$-160 I_m [\sin 5000t \cos \theta + \cos 5000t \sin \theta] + 90 I_m [\cos 5000t \cos \theta - \sin 5000t \sin \theta] - 1750 [\cos 5000t \cos 30^\circ - \sin 5000t \sin 30^\circ] = 0$$

$$[-160 I_m \cos \theta - 90 I_m \sin \theta + 1750 \sin 30^\circ] \sin 5000t + [-160 I_m \sin \theta + 90 I_m \cos \theta - 1750 \cos 30^\circ] \cos 5000t = 0$$

$$\sin 30^\circ = 0,5 \quad \cos 30^\circ = \frac{\sqrt{3}}{2}$$

②



$$\therefore +160 I_m \cos \theta + 90 I_m \sin \theta = 750 \times 0,5 = 375 \dots (*)$$

$$-160 I_m \sin \theta + 90 I_m \cos \theta = 750 \times \frac{\sqrt{3}}{2} = 375\sqrt{3} \dots (**)$$

$$\begin{aligned} & \rightarrow 25600 I_m^2 \cos^2 \theta + 8100 I_m^2 \sin^2 \theta + 28800 I_m^2 \cos \theta \sin \theta \\ & = 140625 \end{aligned}$$

$$\begin{aligned} +) & 25600 I_m^2 \sin^2 \theta + 8100 I_m^2 \cos^2 \theta - 28800 I_m^2 \sin \theta \cos \theta \\ & = 421875 \end{aligned}$$

$$\begin{aligned} & \frac{25600 I_m^2 (\underbrace{\cos^2 \theta + \sin^2 \theta}_{=1}) + 8100 I_m^2 (\underbrace{\sin^2 \theta + \cos^2 \theta}_{=1})}{=1} \\ & = 562500 \end{aligned}$$

$$\therefore 25600 I_m^2 + 8100 I_m^2 = 562500$$

$$33700 I_m^2 = 562500$$

$$\therefore I_m^2 = 16,69139$$

$$\therefore I_m = 4,0855$$

(3)

$$(*) \quad 160 I_m \cos \theta + 90 I_m \sin \theta = 375 \quad \text{--- (a)}$$

$$(**) \quad -160 I_m \sin \theta + 90 I_m \cos \theta = 375\sqrt{3}$$



$$90 I_m \cos \theta - 160 I_m \sin \theta = 375\sqrt{3} \quad \text{--- (b)}$$

$$(a) \times 9$$

$$1440 I_m \cos \theta + 810 I_m \sin \theta = 3375$$

$$(b) \times 16$$

$$1440 I_m \cos \theta - 2560 I_m \sin \theta = 6000\sqrt{3} \quad \text{---}$$

$$3370 I_m \sin \theta = -17017.305$$

$$\therefore \sin \theta = \frac{-17017.305}{3370 \times 4.0855 I_m}$$

$$\therefore \theta = -0.509677$$

$$\therefore \theta = \sin^{-1}(-0.509677)$$

$$= -30.6423^\circ$$

$$\therefore i(t) = I_m \cos(5000t + \theta) \\ = 4.0855 \cos(5000t - 30.6423^\circ)$$

§9.3 The Phasor

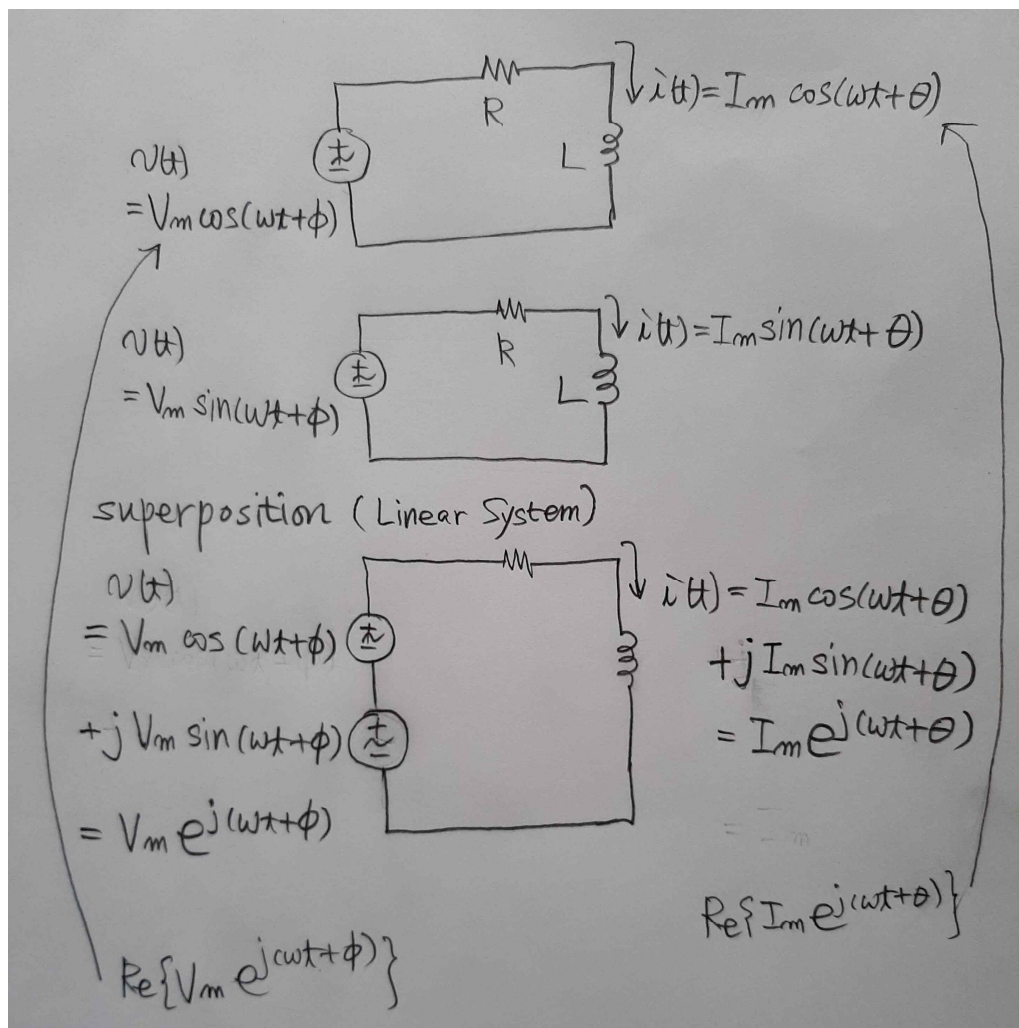
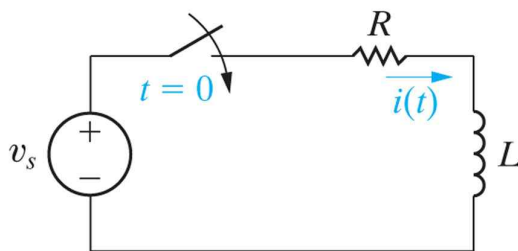
- Euler's Identity

$$e^{\pm j\theta} = \cos \theta \pm j \sin \theta$$

$$\cos \theta = \operatorname{Re}\{e^{j\theta}\} \text{ and } \sin \theta = \operatorname{Im}\{e^{j\theta}\}$$

(※ $e^{j\pi} + 1 = 0$: Some mathematicians say this equation is the most beautiful eq. in the world, since it includes e , π , 0, 1 and j .)

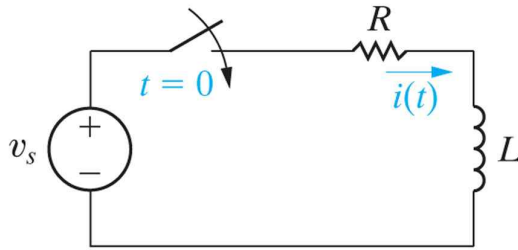
- Input and output of a Linear System



- Summary

input of a linear system	output of the system
$v(t) = V_m \cos(\omega t + \phi)$	$i(t) = I_m \cos(\omega t + \theta)$
$v(t) = V_m \sin(\omega t + \phi)$	$i(t) = I_m \sin(\omega t + \theta)$
$v(t) = V_m \cos(\omega t + \phi)$ $+ j V_m \sin(\omega t + \phi)$ $= V_m e^{j(\omega t + \phi)}$	$i(t) = I_m \cos(\omega t + \theta)$ $+ j I_m \sin(\omega t + \theta)$ $= I_m e^{j(\omega t + \theta)}$
$v(t) = V_m \cos(\omega t + \phi)$ $= \operatorname{Re} \{ V_m e^{j(\omega t + \phi)} \}$	$i(t) = I_m \cos(\omega t + \theta)$ $= \operatorname{Re} \{ I_m e^{j(\omega t + \theta)} \}$

● RL circuit with Euler's Identity



$$v_s(t) = V_m e^{j(\omega t + \phi)}, \quad \text{Let } i(t) = I_m e^{j(\omega t + \theta)}$$

$$L \frac{di(t)}{dt} + R i(t) = v_s(t)$$

$$L \frac{d}{dt} I_m e^{j(\omega t + \theta)} + R I_m e^{j(\omega t + \theta)} = V_m e^{j(\omega t + \phi)}$$

$$j\omega L I_m e^{j(\omega t + \theta)} + R I_m e^{j(\omega t + \theta)} = V_m e^{j(\omega t + \phi)}$$

$$[j\omega L + R] I_m e^{j\theta} e^{j\omega t} = V_m e^{j\phi} e^{j\omega t}$$

$$[j\omega L + R] I_m e^{j\theta} = V_m e^{j\phi} \quad (*)$$

$$I_m e^{j\theta} = \frac{V_m}{R + j\omega L} e^{j\phi}$$

$$I_m e^{j\theta} = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} e^{j[\phi - \tan^{-1}(\omega L/R)]}$$

$$\therefore I_m = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}}, \quad \theta = \phi - \tan^{-1}(\omega L/R)$$

$$\therefore i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos \left[\omega t + \phi - \tan^{-1} \left(\frac{\omega L}{R} \right) \right]$$

※ After eliminating $e^{j\omega t}$ from both sides, we can get Eq.(*).

※ Eq.(*) : NO time (t) terms in the equation.

This equation includes not time (t) but frequency (ω).

From now our playground is the frequency domain,

not the time domain. (a very important concept).

From now we will analyze a circuit in the frequency domain.

● Phasor transform

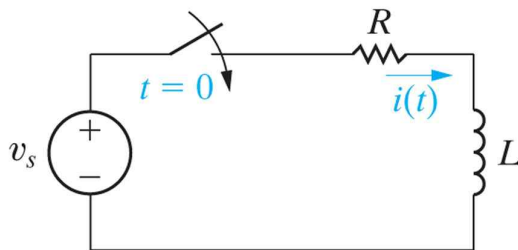
time domain representation

$$\begin{aligned}
 v(t) &= V_m \cos(\omega t + \phi) \\
 &= V_m \operatorname{Re}\{e^{(j\omega t + \phi)}\} \\
 &= V_m \operatorname{Re}\{e^{j\omega t} e^{j\phi}\} \\
 &= \operatorname{Re}\{V_m e^{j\phi} e^{j\omega t}\}
 \end{aligned}$$

frequency domain representation

$$\begin{aligned}
 \mathbf{V} &= P\{V_m \cos(\omega t + \phi)\} \quad (\text{* phasor transform}) \\
 &= V_m e^{j\phi} = V_m \angle \phi \quad (\text{PHASOR [* important concept]}) \\
 &= V_m (\cos \phi + j \sin \phi) \quad (\text{* Phasors are COMPLEX numbers.})
 \end{aligned}$$

● RL circuit with the phasor



Reuse Eq.(*)

$$[R + j\omega L] I_m e^{j\theta} = V_m e^{j\phi}$$

$$\text{let } \mathbf{I} = I_m e^{j\theta} = I_m \angle \theta \quad (\text{phasor})$$

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi \quad (\text{phasor})$$

$$[R + j\omega L] I_m \angle \theta = V_m \angle \phi \quad \text{or} \quad [R + j\omega L] \mathbf{I} = \mathbf{V}$$

$$I_m \angle \theta = \frac{V_m \angle \phi}{R + j\omega L} \quad \text{or} \quad \mathbf{I} = \frac{\mathbf{V}}{R + j\omega L}$$

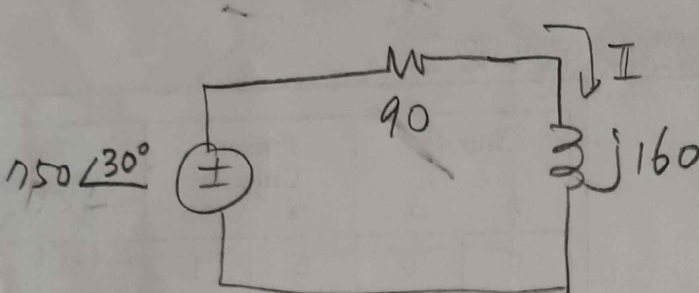
$$\text{let } \mathbf{Z} = R + j\omega L \quad (\text{Impedance})$$

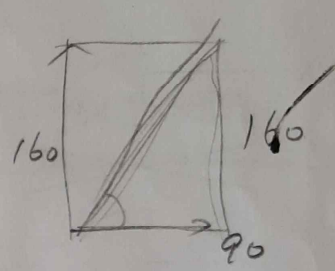
$$\mathbf{I} = \frac{\mathbf{V}}{\mathbf{Z}} \quad (\text{Ohm's Law})$$

(Example) Phasor method

④

(b)



$$\begin{aligned} \underline{I} &= \frac{750 \angle 30^\circ}{90 + j160} \\ &= \frac{750 \angle 30^\circ}{\sqrt{8100 + 25600} \angle 60.6422^\circ} \\ &= \frac{750 \angle 30^\circ}{183.51156 \angle 60.6422^\circ} \\ &= 4.0855 \angle -30.6422^\circ \end{aligned}$$


$$\therefore i(t) = 4.0855 \cos(5000t - 30.6422^\circ)$$

● time domain and frequency domain

time domain	frequency domain (phasor)
$v_s(t) = V_m \cos(\omega t + \phi)$	$\mathbf{V} = V_m e^{j\phi} = I_m \angle \phi$
$i(t) = I_m \cos(\omega t + \theta)$	$\mathbf{I} = I_m e^{j\theta} = I_m \angle \theta$
$R, L, d/dt$	$R, L, j\omega$
$L \frac{di(t)}{dt} + Ri(t) = v_s(t)$	$(R + j\omega L) I_m \angle \theta = V_m \angle \phi$
differential equation	algebraic equation, but complex numbers
.....	$I_m \angle \theta = \frac{V_m \angle \phi}{R + j\omega L}$
$i(t) = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \cos[\omega t + \phi - \tan^{-1}(\frac{\omega L}{R})]$	

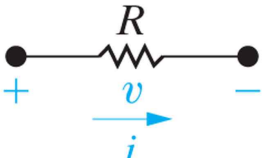
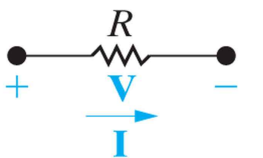
(Example) $y_1(t) = 20 \cos(\omega t - 30^\circ)$, $y_2(t) = 40 \cos(\omega t + 60^\circ)$

$y = y_1 + y_2$ (trigonometric method)	$y = y_1 + y_2$ (phasor method)
$y_1(t) = 20 \cos(\omega t - 30^\circ)$ $y_2(t) = 40 \cos(\omega t + 60^\circ)$ using trigonometric theorems $y_1 = 20 \cos \omega t \cos 30^\circ + 20 \sin \omega t \sin 30^\circ$ $y_2 = 40 \cos \omega t \cos 60^\circ - 40 \sin \omega t \sin 60^\circ$ $y = (20 \cos 30^\circ + 40 \cos 60^\circ) \cos \omega t$ $+ (20 \sin 30^\circ - 40 \sin 60^\circ) \sin \omega t$ $= 37.32 \cos \omega t - 24.64 \sin \omega t$ $= 44.72 \left(\frac{37.32}{44.72} \cos \omega t - \frac{24.64}{44.72} \sin \omega t \right)$ $= 44.72 (\cos 33.43^\circ \cos \omega t - \sin 33.43^\circ \sin \omega t)$ $= 44.72 \cos(\omega t + 33.43^\circ)$	$y_1(t) = 20 \cos(\omega t - 30^\circ)$ $y_2(t) = 40 \cos(\omega t + 60^\circ)$ using phasors $\mathbf{Y}_1 = 20 \angle -30^\circ$ $\mathbf{Y}_2 = 40 \angle 60^\circ$ $\mathbf{Y} = \mathbf{Y}_1 + \mathbf{Y}_2$ $= 20 \angle -30^\circ + 40 \angle 60^\circ$ $= (17.32 - j10) + (20 + j34.64)$ $= 37.32 + j24.64$ $= 44.72 \angle 33.43^\circ$ $\therefore y = 44.72 \cos(\omega t + 33.43^\circ)$

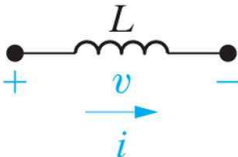
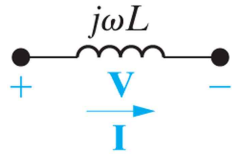
※ The superiority of the phasor approach for adding sinusoidal functions should be apparent. It requires the ability to move back and forth between the polar and rectangular forms of complex numbers.

§9.4 Passive Circuit Elements in the Frequency Domain

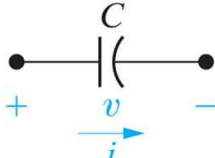
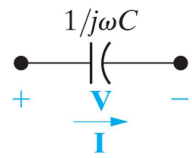
● The V-I Relationship for a Resistor

time domain	frequency domain [phasor]
	
$i(t) = I_m \cos(\omega t + \theta_i)$ $v(t) = R I_m \cos(\omega t + \theta_i)$ $\therefore v(t) = R i(t)$	$I = I_m \angle \theta_i$ $V = R I_m \angle \theta_i$ $\therefore V = R I$

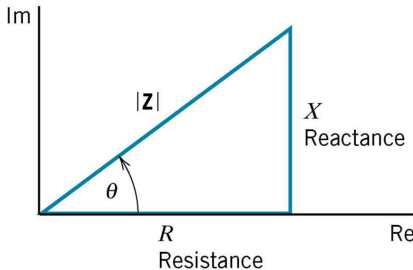
● The V-I Relationship for an Inductor

time domain	frequency domain [phasor]
	
$i(t) = I_m \cos(\omega t + \theta_i)$ $v(t) = L \frac{di(t)}{dt}$ $= -\omega L I_m \cos(\omega t + \theta_i - 90^\circ)$	$I = I_m \angle \theta_i$ $V = j\omega L (I_m \angle \theta_i)$ $\therefore V = (j\omega L) I$ (a kind of the Ohm's law)

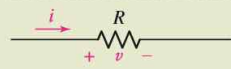
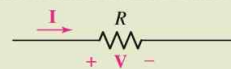
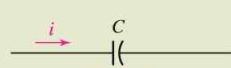
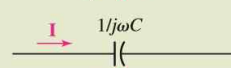
● The V-I Relationship for a Capacitor

time domain	frequency domain [phasor]
	
$i(t) = C \frac{dv(t)}{dt}$ $= -\omega C V_m \cos(\omega t + \theta_v - 90^\circ)$	$V = V_m \angle \theta_v$ $I = j\omega C (V_m \angle \theta_v)$ $\therefore I = (j\omega C) V$ (a kind of the Ohm's law)

● Impedance

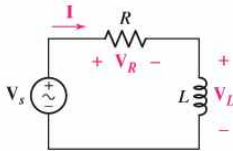
resistor $V = RI$	Impedance Z (unit: Ω) $V = ZI$ 
inductor $V = (j\omega L) I$	
capacitor $V = \frac{1}{j\omega C} I$ $= -j \frac{1}{\omega C} I$	

(Summary) Comparison of Time-Domain and Frequency-Domain Voltage-Current Expressions

Time Domain		Frequency Domain	
	$v = Ri$	$V = RI$	
	$v = L \frac{di}{dt}$	$V = j\omega LI$	
	$v = \frac{1}{C} \int i dt$	$V = \frac{1}{j\omega C} I$	

§9.5 Kirchhoff's Laws in the Frequency Domain

	time domain	frequency domain
KVL	$\sum_{n=1}^N v_n(t) = 0$	$\sum_{n=1}^N V_n = 0$
KCL	$\sum_{n=1}^N i_n(t) = 0$	$\sum_{n=1}^N I_n = 0$
	Real number equation, complex calculation	complex number equation, easy calculation



■ **FIGURE 10.16** The series RL circuit with a phasor voltage applied.

Now let us look briefly at the series RL circuit that we have considered several times before. The circuit is shown in Fig. 10.16, and a phasor current and several phasor voltages are indicated. We may obtain the desired response, a time-domain current, by first finding the phasor current. From Kirchhoff's voltage law,

$$\mathbf{V}_R + \mathbf{V}_L = \mathbf{V}_s$$

and using the recently obtained $\mathbf{V}$ - $\mathbf{I}$ relationships for the elements, we have

$$\mathbf{R}\mathbf{I} + j\omega L\mathbf{I} = \mathbf{V}_s$$

The phasor current is then found in terms of the source voltage $\mathbf{V}_s$:

$$\mathbf{I} = \frac{\mathbf{V}_s}{R + j\omega L}$$

Let us select a source-voltage amplitude of V_m and phase angle of 0° . Thus,

$$\mathbf{I} = \frac{V_m \angle 0^\circ}{R + j\omega L}$$

The current may be transformed to the time domain by first writing it in polar form:

$$\mathbf{I} = \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \angle [-\tan^{-1}(\omega L/R)]$$

and then following the familiar sequence of steps to obtain in a very simple manner the same result we obtained the “hard way” earlier in this chapter.

Example 9.6 KVL in the Frequency Domain (Textbook p. 321)

A $90\ \Omega$ resistor, a $32\ \text{mH}$ inductor, and a $5\ \mu\text{F}$ capacitor are connected in series across the terminals of a sinusoidal voltage source, as shown in Fig. 9.15. The steady-state expression for the source voltage v_s is $750 \cos(5000t + 30^\circ)\ \text{V}$.

- Construct the frequency-domain equivalent circuit.
- Calculate the steady-state current i by the phasor method.

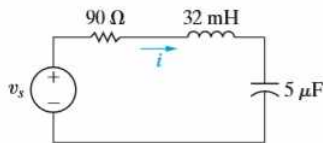


Figure 9.15 ▲ The circuit for Example 9.6.

Solution

- From the expression for v_s , we have $\omega = 5000\ \text{rad/s}$. Therefore the impedance of the $32\ \text{mH}$ inductor is

$$Z_L = j\omega L = j(5000)(32 \times 10^{-3}) = j160\ \Omega,$$

and the impedance of the capacitor is

$$Z_C = j \frac{-1}{\omega C} = -j \frac{10^6}{(5000)(5)} = -j40\ \Omega.$$

The phasor transform of v_s is

$$\mathbf{V}_s = 750 \angle 30^\circ\ \text{V}.$$

Figure 9.16 illustrates the frequency-domain equivalent circuit of the circuit shown in Fig. 9.15.

- We compute the phasor current simply by dividing the voltage of the voltage source by the equivalent impedance between the terminals a,b. From Eq. 9.45,

$$\begin{aligned} Z_{ab} &= 90 + j160 - j40 \\ &= 90 + j120 = 150 \angle 53.13^\circ\ \Omega. \end{aligned}$$

Thus

$$\mathbf{I} = \frac{750 \angle 30^\circ}{150 \angle 53.13^\circ} = 5 \angle -23.13^\circ\ \text{A}.$$

We may now write the steady-state expression for i directly:

$$i = 5 \cos(5000t - 23.13^\circ)\ \text{A}.$$

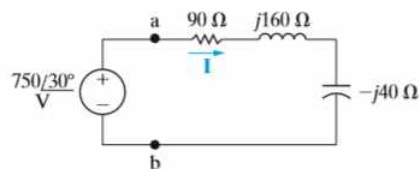
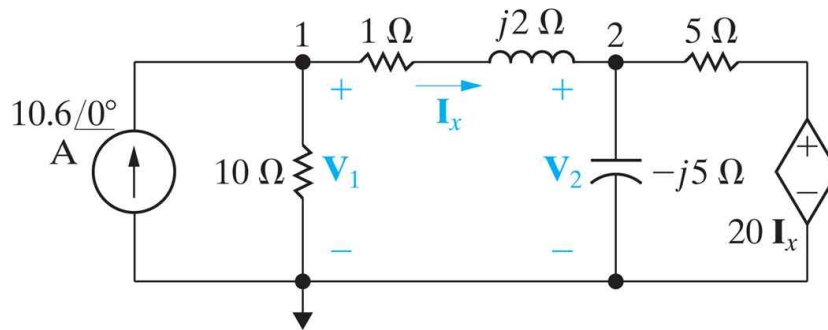


Figure 9.16 ▲ The frequency-domain equivalent circuit of the circuit shown in Fig. 9.15.

§9.7 The Node-Voltage Method in Frequency Domain



$$-10.6 + \frac{V_1}{10} + \frac{V_1 - V_2}{1 + j2} = 0$$

$$\frac{V_2 - V_1}{1 + j2} + \frac{V_2}{-j5} + \frac{V_2 - 20I_x}{5} = 0$$

$$I_x = \frac{V_1 - V_2}{1 + j2}$$

Solving the above 3 complex number equations, we can get

$$\therefore V_1 = 68.40 - j16.80 \text{ [V]}$$

$$V_2 = 68 - j26 \text{ [V]}$$

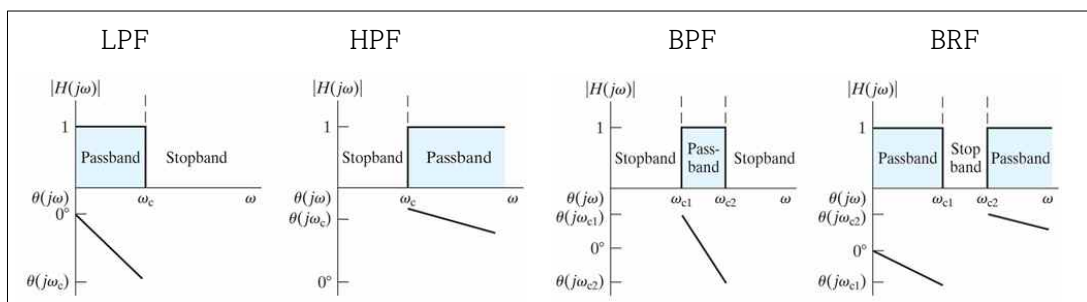
$$I_x = 3.76 + j1.68 \text{ [A]}$$

Chap.14 Frequency Selective Circuits [Filters]

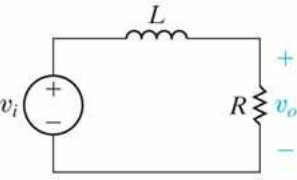
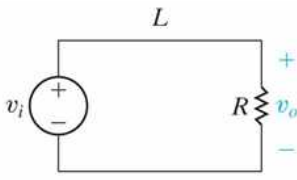
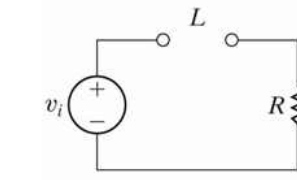
- Frequency selective circuits are also called FILTERs.



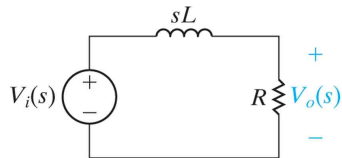
§14.1 Frequency Responses of Ideal Filters



§14.2 Low-Pass Filters (LPF)

a series RL LPF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)
		

- The series RL circuit – Quantitative Analysis



the frequency response

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{R}{j\omega L + R} = \frac{R/L}{j\omega + R/L} \quad \text{Eq.(\#)}$$

the magnitude plot and the angle plot of the frequency response

$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}}, \quad \theta(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

※ Defining the Cutoff Frequency (ω_c or f_c)

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} H_{\max}$$

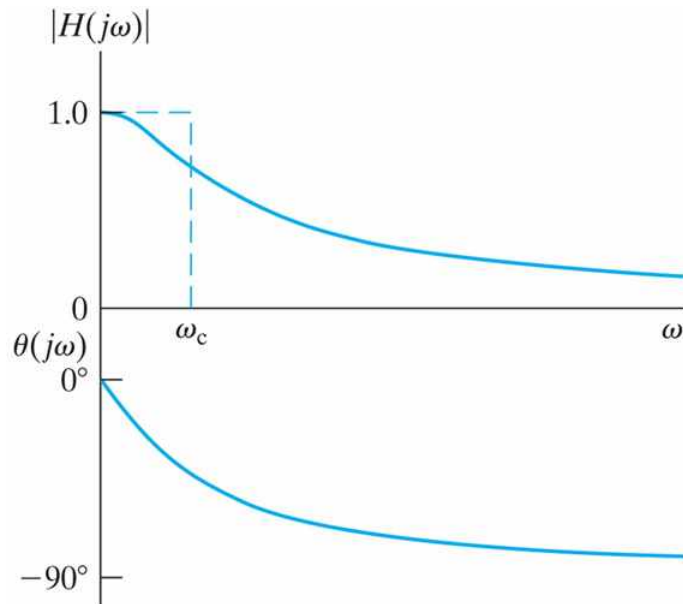
the cutoff frequency of the series RL circuit

$$H_{\max} = |H(j0)| = 1$$

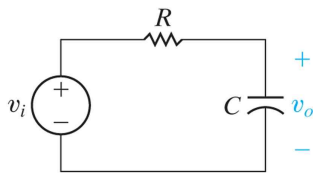
$$|H(j\omega_c)| = \frac{R/L}{\sqrt{\omega_c^2 + (R/L)^2}} = \frac{1}{\sqrt{2}} |1| \quad \therefore \omega_c = \frac{R}{L}$$

※ time constant : $\tau = \frac{L}{R}$

- Frequency response of an RL LPF



- A series RC circuit



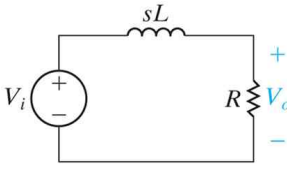
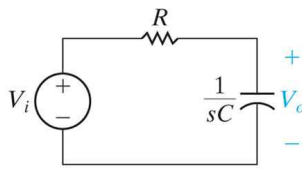
the frequency response

$$H(j\omega) = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{1}{j\omega RC + 1} = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}} \quad \text{Eq. (##)}$$

(time constant : $\tau = RC$)

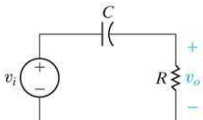
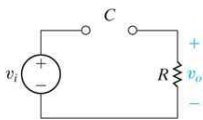
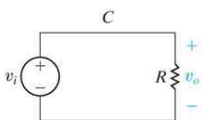
cutoff frequency : $\omega_c = \frac{1}{RC}$

※ Comparison [Low Pass Filters]

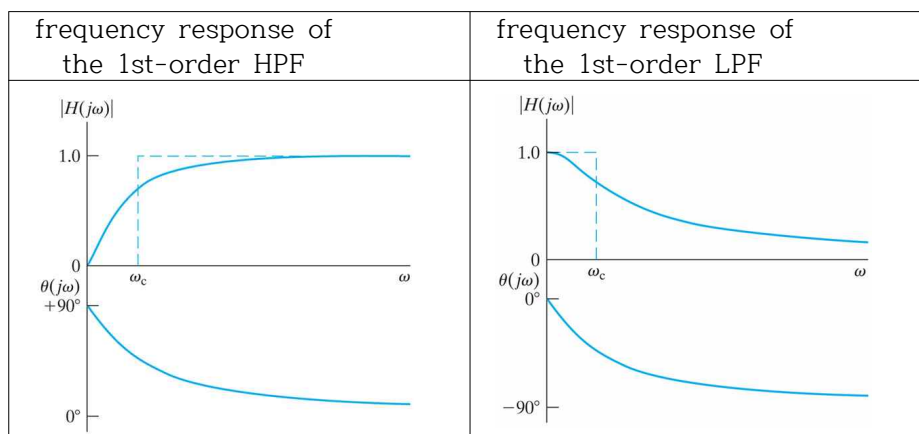
equation	Eq.(#)	Eq.##)
RL or RC	RL circuit	RC circuit
circuit		
Frequency Response	$H(j\omega) = \frac{R/L}{j\omega + R/L}$	$H(j\omega) = \frac{\frac{1}{RC}}{j\omega + \frac{1}{RC}}$
Time Constant	$\tau = \frac{L}{R} \quad [\text{sec}]$	$\tau = RC \quad [\text{sec}]$
Cutoff Frequency	$\omega_c = \frac{R}{L} \quad [\text{rad/sec}]$ (or $f_c = \frac{1}{2\pi} \left(\frac{R}{L} \right) \quad [\text{Hz}]$)	$\omega_c = \frac{1}{RC} \quad [\text{rad/sec}]$ (or $f_c = \frac{1}{2\pi} \left(\frac{1}{RC} \right) \quad [\text{Hz}]$)
Remark	$H(j\omega) = \frac{\text{constant}}{j\omega + \text{constant}} = \frac{\omega_c}{j\omega + \omega_c}$ 1. General form of first-order LPF 2. one memory circuits 3. first-order differential equation in time domain 4. frequency response in frequency domain	

§14.3 High-Pass Filters (HPF)

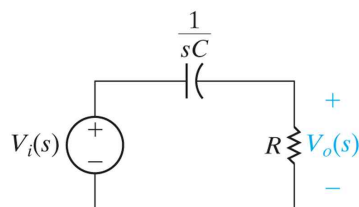
- A series RC circuit

a series RC HPF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)
		

- Frequency response of the first-order RC HPF



- the first-order RC HPF



the frequency response

$$H(j\omega) = \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{j\omega}{j\omega + 1/RC} \quad \text{Eq. (*)}$$

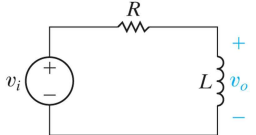
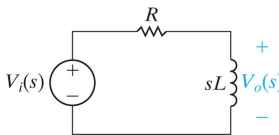
the magnitude and the angle of the frequency response

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + (1/RC)^2}} \quad , \quad \theta(j\omega) = 90^\circ - \tan^{-1} \omega RC$$

the cutoff frequency : $\omega_c = \frac{1}{RC}$ (* time constant : $\tau = RC$)

* Same cutoff frequency (or time constant) with LPF

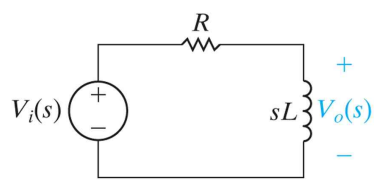
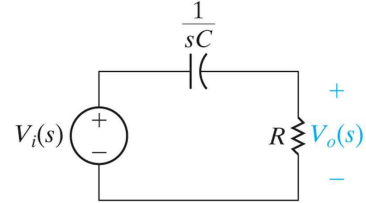
- (example) A series RL HPF

time-domain	frequency-domain
	

frequency response : $H(j\omega) = \frac{j\omega}{j\omega + R/L}$ Eq.(**)

cutoff frequency : $\omega_c = \frac{R}{L}$

※ Comparison [High Pass Filters]

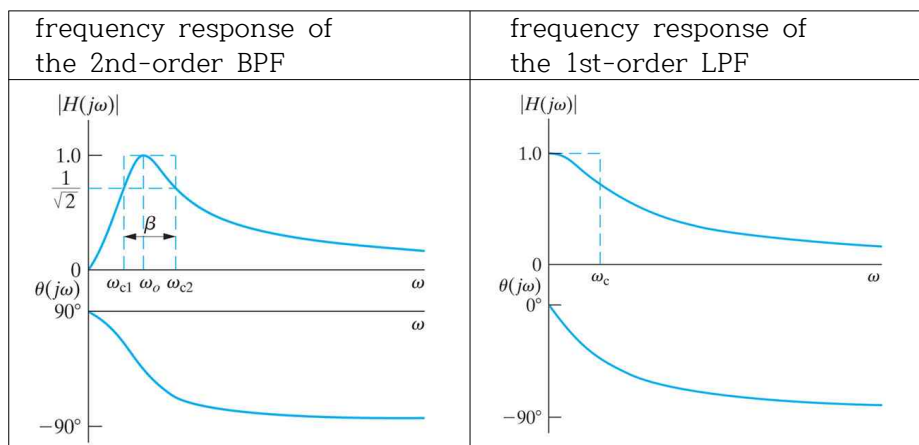
equation	Eq.(**)	Eq.(*)
RL or RC	RL circuit	RC circuit
circuit		
Frequency Response	$H(j\omega) = \frac{j\omega}{j\omega + R/L}$	$H(j\omega) = \frac{j\omega}{j\omega + 1/RC}$
Time Constant	$\tau = \frac{L}{R}$ [sec]	$\tau = RC$ [sec]
Cutoff Frequency	$\omega_c = \frac{R}{L}$ [rad/sec] (or $f_c = \frac{1}{2\pi} \left(\frac{R}{L} \right)$ [Hz])	$\omega_c = \frac{1}{RC}$ [rad/sec] (or $f_c = \frac{1}{2\pi} \left(\frac{1}{RC} \right)$ [Hz])
Remark	$H(j\omega) = \frac{j\omega}{j\omega + \text{constant}} = \frac{j\omega}{j\omega + \omega_c}$ - General form of first-order HPF - one memory circuits - first-order differential equation in time domain - frequency response in frequency domain	

§14.4 Bandpass Filters (BPF)

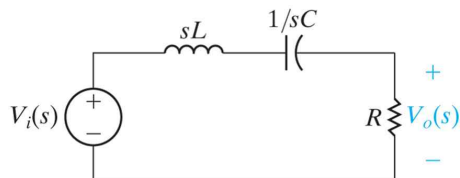
- A series RLC bandpass filter

a series RLC BPF	equivalent circuit ($\omega = 0$)	equivalent circuit ($\omega = \infty$)

- Frequency response of the second-order RLC BPF



The series RLC Circuit



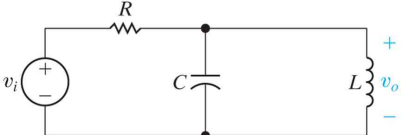
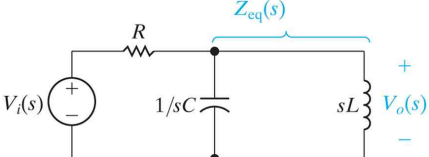
- frequency response

$$\begin{aligned}
 H(j\omega) &= \frac{V_o(j\omega)}{V_i(j\omega)} = \frac{R}{R + j\omega L + \frac{1}{j\omega C}} = \frac{R}{R + j(\omega L - \frac{1}{\omega C})} \\
 &= \frac{(R/L)(j\omega)}{(j\omega)^2 + (R/L)(j\omega) + (1/LC)}
 \end{aligned}$$

$$\text{(magnitude)} \quad |H(j\omega)| = \frac{\omega(R/L)}{\sqrt{[(1/LC) - \omega^2]^2 + [\omega(R/L)]^2}}$$

$$\text{(phase)} \quad \theta(j\omega) = 90^\circ - \tan^{-1} \left[\frac{\omega(R/L)}{(1/LC) - \omega^2} \right]$$

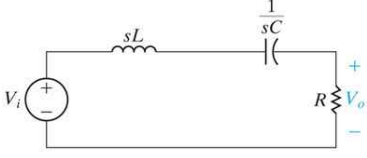
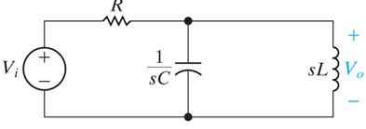
- a parallel RLC BPF

a parallel RLC BPF	the frequency-domain equivalent
	

- frequency response

$$H(j\omega) = \frac{\frac{j\omega}{RC}}{j\omega^2 + \frac{j\omega}{RC} + \frac{1}{LC}}$$

- Bandpass Filters

a series RLC BPF	a parallel RLC BPF
 $H(s) = \frac{(R/L)s}{s^2 + (R/L)s + 1/LC}$ $\omega_o = \sqrt{1/LC} \quad \beta = R/L$	 $H(s) = \frac{s/RC}{s^2 + s/RC + 1/LC}$ $\omega_o = \sqrt{1/LC} \quad \beta = 1/RC$

- General form of the frequency response for BPF

$$H(j\omega) = \frac{\text{constant} \times (j\omega)}{(j\omega)^2 + \text{constant} \times (j\omega) + \text{constant}}$$

$$= \frac{\beta(j\omega)}{(j\omega)^2 + \beta(j\omega) + \omega_0^2}$$

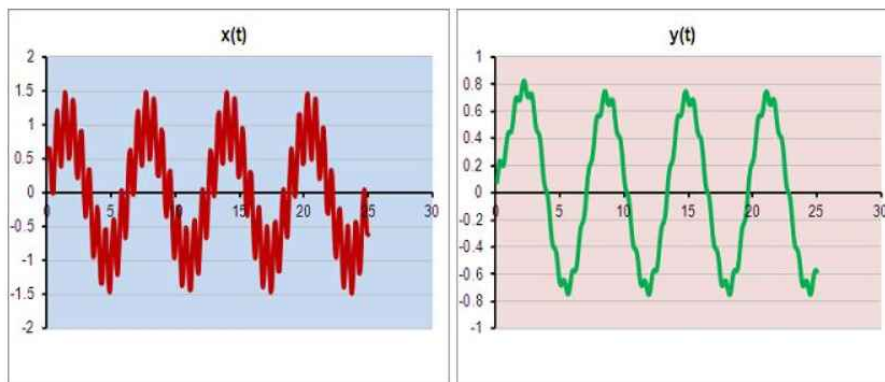
$$\text{where } \omega_0 = \sqrt{\frac{1}{LC}}$$

$$\text{and } \beta = \frac{R}{L} \quad (\text{series circuit}) \quad \text{or} \quad \beta = \frac{1}{RC} \quad (\text{parallel circuit})$$

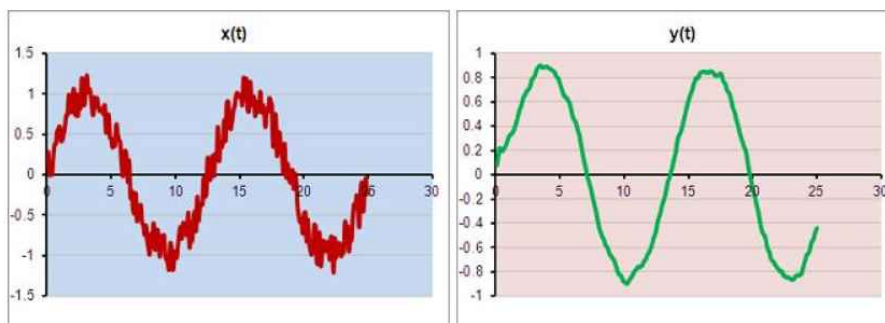
- Summary

LPF	$H(j\omega) = \frac{\text{constant}}{(j\omega) + \text{constant}}$
HPF	$H(j\omega) = \frac{(j\omega)}{(j\omega) + \text{constant}}$
BPF	$H(j\omega) = \frac{\text{constant} \times (j\omega)}{(j\omega)^2 + \text{constant} \times (j\omega) + \text{constant}}$
BRF	$H(j\omega) = \frac{(j\omega)^2 + \text{constant}}{(j\omega)^2 + \text{constant} \times (j\omega) + \text{constant}}$

(Example) LPF

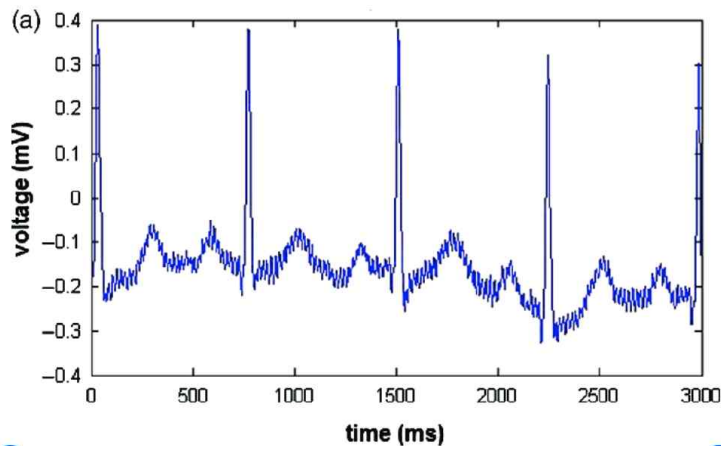


The filtration can be made better by using a larger filter time constant (τ), how tradeoff between the amount of filtering and attenuation of the desired signal. An e also helps in filtering out random noise from a signal, see the effect below on i modelled by $\text{SIN}(0.5 \times x) + 0.5 * (\text{RAND}() - 0.5)$

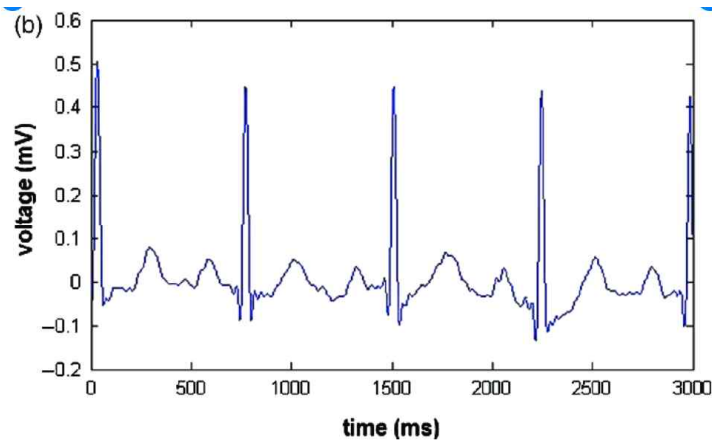


(Example) ECG (ElectroCardioGraph)

(a) measured signal



(b) filtered signal

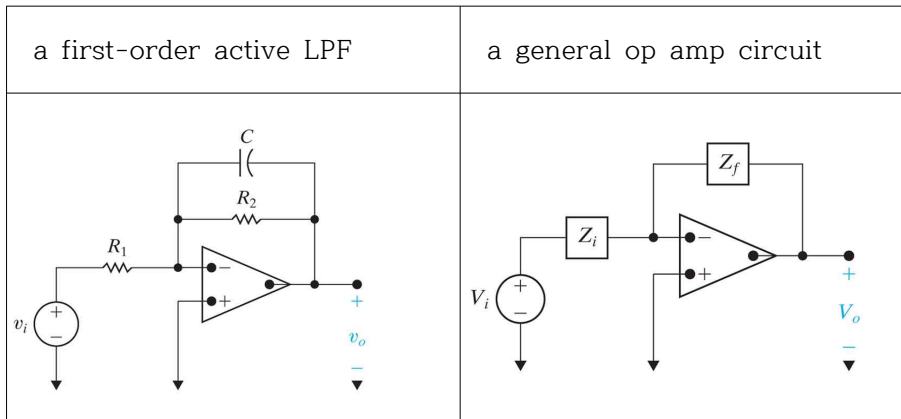


period $T = 0.75$ sec

heart rate = 80/minute

Chap.15 Active Filter Circuits

§15.1 First-Order Active Low-Pass Filter



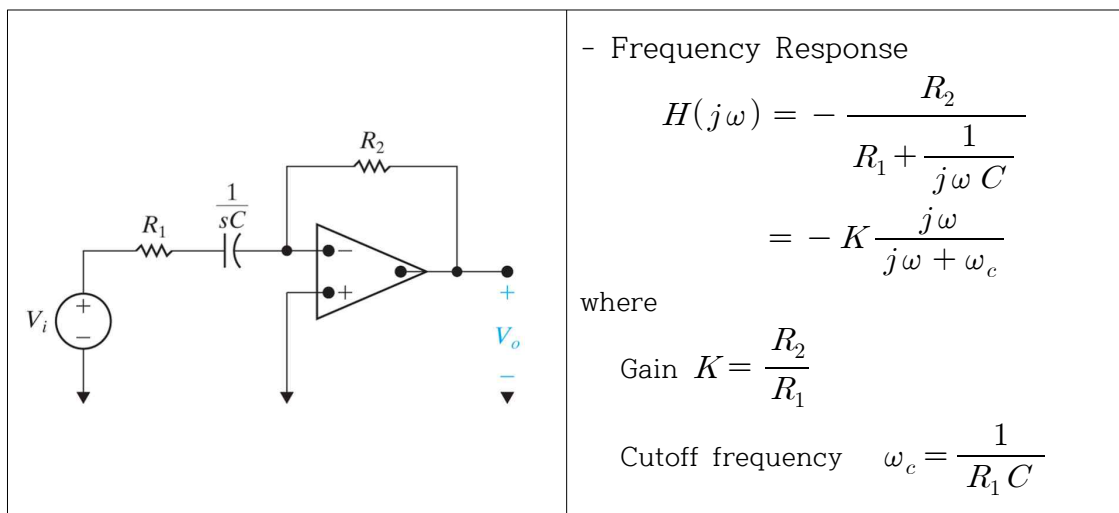
- Frequency Response

$$\begin{aligned}
 H(j\omega) &= -\frac{Z_f}{Z_i} = -\frac{R_2 \parallel \left(\frac{1}{j\omega C}\right)}{R_1} \\
 &= -\frac{R_2}{R_1} \frac{\frac{1}{R_2 C}}{j\omega + \frac{1}{R_2 C}} \equiv -K \frac{\omega_c}{j\omega + \omega_c}
 \end{aligned}$$

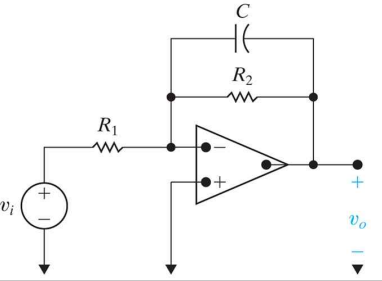
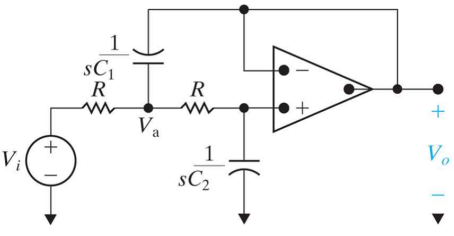
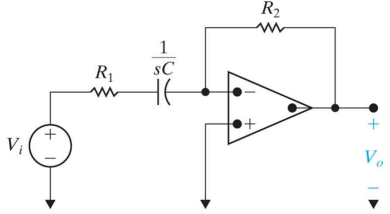
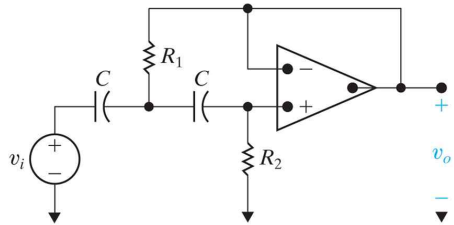
K : gain, ω_c : cutoff frequency

※ same type with Eq.(#) or Eq.(##), that is, Low Pass Filter

● First-Order Active High-Pass Filter



● Butterworth Filter

first-order LPF	second-order LPF
	
$H(j\omega) = -\frac{R_2}{R_1} \frac{\frac{1}{R_2 C}}{j\omega + \frac{1}{R_2 C}}$	$H(j\omega) = \frac{1}{R^2 C_1 C_2} \frac{1}{(j\omega)^2 + \frac{2}{R C_1} (j\omega) + \frac{1}{R^2 C_1 C_2}}$
first-order HPF	second-order HPF
	
$H(j\omega) = -\frac{R_2}{R_1} \frac{(j\omega)}{(j\omega) + \frac{1}{R_1 C}}$	$H(j\omega) = \frac{(j\omega)^2}{(j\omega)^2 + \frac{2}{R_2 C} (j\omega) + \frac{1}{R_1 R_2 C^2}}$